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The Probability of Neutron Decay and its Effects on Radioactivity

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Abstract

Neutrons play a crucial role in the decay of atoms, exerting direct influence over radioactive elements with wide-ranging implications in nuclear physics and practical applications like nuclear waste management and medical diagnostics. This study delves into the dynamics of free neutron decay, examining its variability in transitioning to protons and proposing a new structural model of the neutron and decay process that aligns with recent observations. Examining the subtle distinctions in decay times observed across different experimental methodologies, such as the bottle and beam methods, revealed a possible connection between neutron decay rates and their interactions with solar neutrinos. A method is proposed to experimentally investigate the acceleration of particle decay processes using neutrino beams, offering new avenues for understanding fundamental nuclear processes.

1. Free Neutron Decay

The neutron, a fundamental particle within the nucleus of an atom, has a pivotal role in maintaining atomic stability. However, outside of the nucleus, in its free state, the neutron is inherently unstable and undergoes decay. This phenomenon is of immense importance in understanding the broader concept of radioactivity.

When a neutron exists outside the nucleus, it is referred to as a free neutron. Unlike its stable state within certain atomic nuclei, a free neutron is susceptible to decay. It undergoes a transformation into a proton, accompanied by the emission of an electron and an antineutrino. The average time taken for a free neutron to decay, as determined by the bottle method, is **approximately 14 minutes and 39 seconds**.¹ This decay process is currently believed to be governed by the weak nuclear force, which mediates interactions between particles and is responsible for processes such as beta decay.

The probability of neutron decay is a fundamental aspect of particle physics. It is characterized by a probability function, dictating the likelihood of a neutron undergoing decay within a given timeframe. While the free neutron decay process and timeframe is relatively well understood, variations in decay times exist among different atomic elements – when the neutron is not *free* and embedded in an atom's nucleus. These differences in decay rates are fundamental to our understanding of nuclear physics and are often quantified in terms of half-life — the time it takes for half of a sample of radioactive material to decay into more stable forms.

The variation in decay times among different atomic elements underscores the intricate nature of nuclear processes. Each element has its own unique set of properties, including particle mass and nuclear structure, which influence the likelihood and speed of decay. For instance, elements with larger atomic masses tend to have longer half-lives, as the

greater number of nucleons provides more stability to the nucleus. Conversely, elements with smaller atomic masses may exhibit shorter half-lives, as their nuclei are more prone to instability and decay.

One prominent example of varying decay times is observed in the uranium isotopes. Uranium-238, a naturally occurring isotope, has a half-life of approximately 4.5 billion years. In contrast, uranium-235, another common isotope of uranium, has a significantly shorter half-life of about 700 million years. This difference in half-lives is attributed to the subtle variations in nuclear structure, with the former having three additional neutrons compared to the latter.

It will be shown in an upcoming section that the products of the free neutron decay process provide valuable insights into the underlying physics of the process. Upon decay, a **neutron** (n) transforms into a **proton** (p), releasing an **electron** (e) and an **antineutrino** ($\bar{\nu}$).²

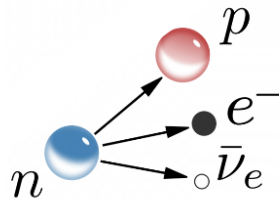


Fig. 1.1 – Neutron decay process
Credit: Wikipedia

2. Variations in Neutron Decay

Despite the accuracy of the bottle method to predict the timing of free neutron decay, there are significant variations in its timing and behavior under different conditions. One of the primary methods for measuring free neutron decay is the bottle method, where ultra-cold neutrons are confined within a container and their decay is monitored over time. However, another common approach, known as the beam method, involves accelerating neutrons to velocities of hundreds of meters per second and detecting the resulting protons after decay. Interestingly, experiments have shown that the **beam method** yields decay times approximately **8-9 seconds longer** than those measured by the bottle method. This discrepancy raises intriguing questions about the influence of neutron velocity on decay probability. Does the higher probability of hitting a stationary target compared to a moving target contribute to the observed differences in decay times?

Another intriguing observation regarding variations in nuclear decay rates was reported during a solar flare on December 13, 2006. Jenkins et al. noted “*a significant decrease in the decay time of ^{54}Mn* ” coinciding with spikes in solar activity.³ While the exact mechanism remains elusive, this finding suggests a potential link between solar phenomena and nuclear decay processes. Could emissions from the Sun, such as neutrinos or other particles, induce shifts in decay rates?

Similarly, studies have revealed slight differences in decay times based on the Earth's orbit around the Sun. An analysis spanning seven years from July 2005 to June 2011 of the decay of ^{36}Cl showed a clear annual variation in decay rates, with maximum counts of decay (*faster decay time*) observed in January/February when the Earth is closer to the Sun and minimum counts of decay (*slower decay time*) in July/August when the Earth is further from the Sun.⁴ This cyclic variation raises intriguing questions about the potential influence of Earth-Sun distance on decay processes.

Further insights into neutron decay variations and distance to the Sun were gained through studies of Venus conducted by the NASA Messenger mission. Wilson et al. investigated neutron decay times above Venus, a planet

closer to the Sun than Earth.⁵ The results of their data analysis suggested a neutron decay time of approximately 780 seconds (13 minutes) on Venus. However, the authors of the paper also added an uncertainty of +/- 130 seconds, making the result more comparable to Earth if the margin of error is considered.

More recently, research by Xiong et al. published in May 2024, provided the most **direct evidence linking neutrinos** to the decay process of neutrons.⁶ They proposed a new nucleosynthesis process, termed the *vr* process, suggesting *“a new nucleosynthesis process that may take place on neutron-rich ejecta experiencing an intensive neutrino flux. The nucleosynthesis proceeds similarly to the standard r process, a sequence of neutron captures and beta decays with, however, charged-current neutrino absorption reactions on nuclei operating much faster than beta decays.”* To reiterate, they found that an intensive neutrino flux on neutrons results in faster beta decay.

The commonality with most of these experiments and variations is the smallest known particle, the neutrino. Produced by the Sun, they are normally emitted with predictability except for solar events like flares. Closer proximity to the Sun should result in a great probability of a neutrino colliding with a neutron. This potential effect of neutrinos and the neutron’s decay process will be explored in an upcoming section.

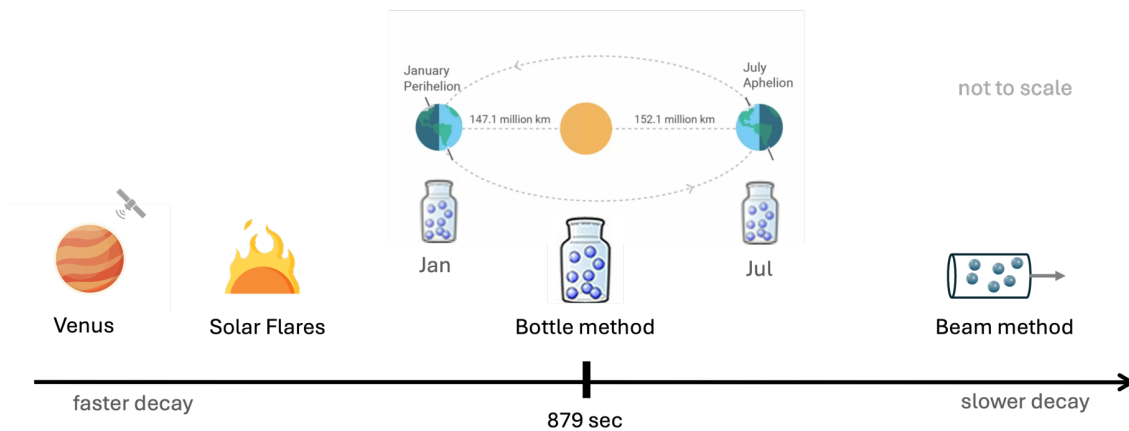


Fig. 2.1 – Decay Variations

3. Current Model of the Neutron - Quarks

The discovery of the neutron marked a pivotal moment in the advancement of our understanding of atomic and subatomic particles. In 1932, Sir James Chadwick, an English physicist, identified the neutron while conducting experiments involving the bombardment of beryllium with alpha particles. This discovery revealed the existence of a particle with mass slightly heavier than that of the proton (by about the additional mass of an electron), yet electrically neutral. It was initially considered to be a single elementary particle.

In 1964, Murray Gell-Mann and George Zweig independently introduced the concept of quarks, postulating that protons and neutrons were composed of smaller constituent particles. The neutron is now known to be a composite particle, not a single elementary particle. According to the three-quark model, the neutron consists of three quarks bound together by the strong nuclear force: two down quarks and one up quark.⁷ This model provided a coherent explanation for the neutron's mass and neutrality, attributing these properties to the specific arrangement and interactions of its constituent quarks.

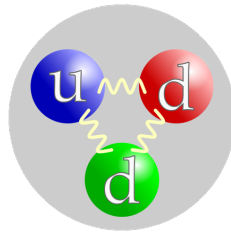


Fig. 3.1 – Neutron 3-Quark Model
Credit: Wikipedia

The proton, like the neutron, is also believed to be composed of three quarks within the framework of the quark model. However, the precise arrangement of quarks within the proton differs from that of the neutron, with two up quarks and one down quark constituting the proton's structure. This arrangement of quarks accounts for the proton's positive charge and slightly lighter mass compared to the neutron.

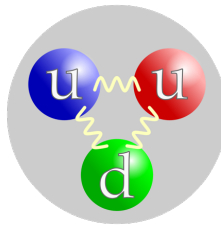


Fig. 3.2 – Proton 3-Quark Model
Credit: Wikipedia

Then in 2015, CERN's discovery of the proton's pentaquark structure, a configuration comprising four quarks and one antiquark, challenged conventional understandings of quark interactions.⁸ Although the proton is still widely regarded as consisting of three quarks, the existence of the pentaquark in newer, high-energy experiments suggests the possibility of exotic configurations beyond the established quark model. Fig. 3.3 illustrates four quarks (black) at the vertices of a tetrahedral structure with a single antiquark (red) at the center.

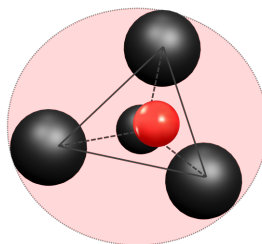


Fig. 3.3 – Proton 5-Quark Model

4. Proposed Model of Neutron

The current understanding of the neutron, as consisting of three quarks bound by the strong nuclear force, has been foundational in shaping our understanding of atomic nuclei and particle interactions. However, recent experimental clues suggest that this model may be incomplete, prompting a reevaluation of the neutron's internal structure and its implications for decay processes.

The first clue challenging the traditional three-quark model stems from advancements in high-energy physics, particularly the discovery of the pentaquark structure within the proton. While the proton was previously believed to consist of three quarks, recent higher energy experiments have revealed evidence of a more complex configuration involving five quarks — a quark-antiquark pair along with the three traditional quarks. If true, it's likely the quark-antiquark pair remained hidden due to their charge cancellation until recently when higher-energy experiments could separate them, like how electrons and positrons can only appear in the pair production process with photon energies that reach gamma ray levels. This revelation of the proton underscores the possibility that the neutron may also contain more than three quarks.

The second clue is the neutron's mass. With a mass nearly equal to that of a proton and an electron combined, the neutron's mass poses a puzzle within the context of the three-quark model. Why is the mass of the neutron = proton + electron?

The third clue is the decay process of the neutron. The decay involves the emission of an electron, a phenomenon seemingly at odds with the behavior of quarks. If the neutron consists of quarks, why does it not eject a quark? Where does the electron come from?

The fourth clue is that quarks are never found in isolation. If a quark cannot exist outside of a composite particle and *quark confinement*, what is it? If it is an electron that is ejected from the neutron and not a quark, is it possible that an electron is a quark? And a positron is the antiquark? For this to be true, the mass and charge properties of an electron and positron would need to change when they are within close range of each other. If we stay with this hypothetical and assume their properties can change within “*strong force range*”, the following proposed model of the neutron and proton makes sense. What remains to be answered, and not covered in this paper, is how the electron's properties change when it becomes a quark and vice versa.

Incorporating these insights, the following is the proposed model of the neutron. Building upon the pentaquark structure observed in the proton, the proposed model posits a similar configuration for the neutron, consisting of **four quarks arranged at the vertices of a tetrahedron**, with an **additional quark-antiquark pair at the center**, for a total of six quarks. Like the proton, the tetrahedral quarks do not account for charge and most of their magnetic energy is stored as the gluons that hold the structure together. This arrangement preserves the neutral charge of the neutron while accounting for its mass through the inclusion of an additional electron-turned quark. This model aligns with experimental observations of neutron decay, mass measurements, and recent discoveries in high-energy physics. If correct, the neutron will one day be found to have six quarks (five quarks and an anti-quark). And if all these quarks could be separated and not confined, they would be found to *decay* to five electrons and one positron.

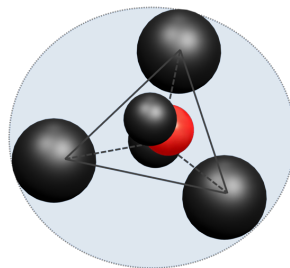


Fig. 4.1 – Neutron 6-Quark Model

5. Probability of Neutron Decay - Neutrinos

Considering the proposed model of the neutron discussed in the previous section, attention is now directed towards explaining the probabilistic nature of neutron decay and its dependence on neutrino interactions. Central to this discussion is the role of neutrinos, elusive particles with the ability to pass through matter virtually undetected due to their minuscule size and interaction with normal matter. Yet, despite their elusive nature, neutrinos play a crucial role in influencing the decay dynamics of free neutrons. Because the neutrino and the antineutrino of the same mass are both neutral particles, it is possible they are the same particle. Thus, this paper refers does not distinguish a difference between the neutrino and antineutrino for the remaining sections.

The probability of neutron decay hinges on the random occurrence of a neutrino striking the center of the neutron. Given the exceedingly small size and weak interaction of neutrinos, a vast number of neutrinos are required to increase the likelihood of collision with a single free neutron. Consider an astonishing 65 billion neutrinos pass through the size of a fingernail every second, with very few, if any, collisions observed. More neutrinos would increase the likelihood of collisions. **The probability of more collisions decreases the lifetime of a neutron.**

Crucially, the source of these neutrinos plays a pivotal role in shaping the probability of neutron decay. Neutrinos emanating from the Sun, for instance, increase the likelihood of neutrino-neutron interactions, particularly for neutrons situated closer to the Sun or during periods of heightened solar activity, such as solar flares. The increased flux of solar neutrinos during such events augments the probability of collisions with free neutrons, accelerating the decay process and influencing the stability of atomic nuclei. This explains the variability seen in experiments during solar flares and with the proximity to the Sun, such as Earth's orbit and the analysis of neutron decay time near Venus. Although it is important to note that the Sun is not the only source of neutrinos that reach Earth.

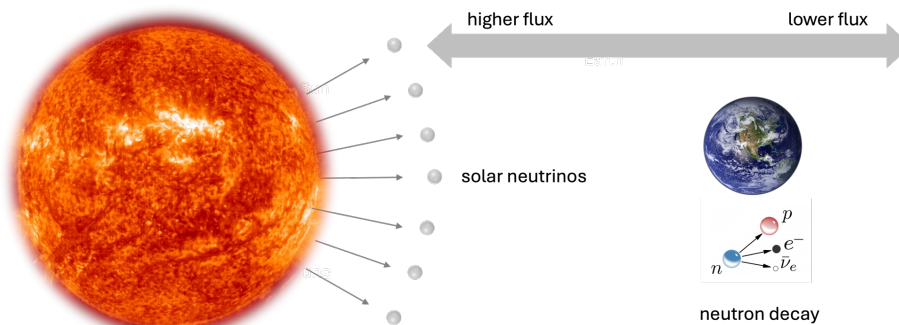


Fig. 5.1 – Solar neutrinos and neutron decay

To explain the difference between the bottle and beam method times for neutron decay, consider the analogy of a bullet striking a stationary car versus a moving car. To illustrate a point, *this example is simplified to one-dimension* to hit the car with an effective width based on the trajectory of a bullet traveling in its direction. Eq. 5.1 is the probability of the bullet hitting the stationary car (P_s), with an effective width of the car (w), possibly hit by a bullet that has a target range along a line length (L). For example, if the total range of the bullets that are fired are a line length of 5, and the car has length of 1, and can be anywhere on this line, then the probability of a single bullet hitting is $P_s = 1/5$. Eq. 5.2 now considers motion of the car. It is similar to the previous equation, but now also considers the velocity that the car is traveling (v) and the time of flight of the bullet (t). Now, it's possible that car is out of range of the original line length (L) by the time the bullet hits, depending on how far the car has traveled while the bullet is in flight. An increase in the velocity of the car increases the denominator of the second equation, thus decreasing the probability of a strike due to the car's motion.

$$P_s = \frac{w}{L}$$

Probability of bullet striking a stationary car

(5.1)

$$P_m = \frac{w}{L + (v \cdot t)}$$

Probability of bullet striking a moving car

(5.2)

When a free neutron is stationary, the probability of a neutrino striking its center and inducing decay is determined by the flux of neutrinos and the size of the target area. However, when the neutron is in motion, as in the beam method of measuring neutron decay, the dynamics of the interaction change, just like the analogy of the bullet and the car. The beam method is often measuring neutrons traveling hundreds of meters per second, decreasing the probability of a strike, and thus increasing the lifetime of a free neutron – by 8 or 9 seconds compared to a stationary measurement using the bottle method.

Now, to summarize and illustrate the neutron's proposed decay process:

1. A random neutrino (ν) strikes the center of the neutron (n). It is a probability function for this randomness, increasing or decreasing probability based on the neutrino flux.
2. If the neutrino strikes the center of the neutron, it emits the center quark (*electron* - e). The antiquark, which is more stable of the two center particles to balance the remaining quarks, remains at the center within the composite structure.
3. After the emission of the center quark (*electron* - e), the neutron becomes a proton (*the pentaquark model of the proton* - p).
4. The decay results match what is found in experiments. A neutron (n) becomes a proton (p), electron (e) and neutrino (ν).

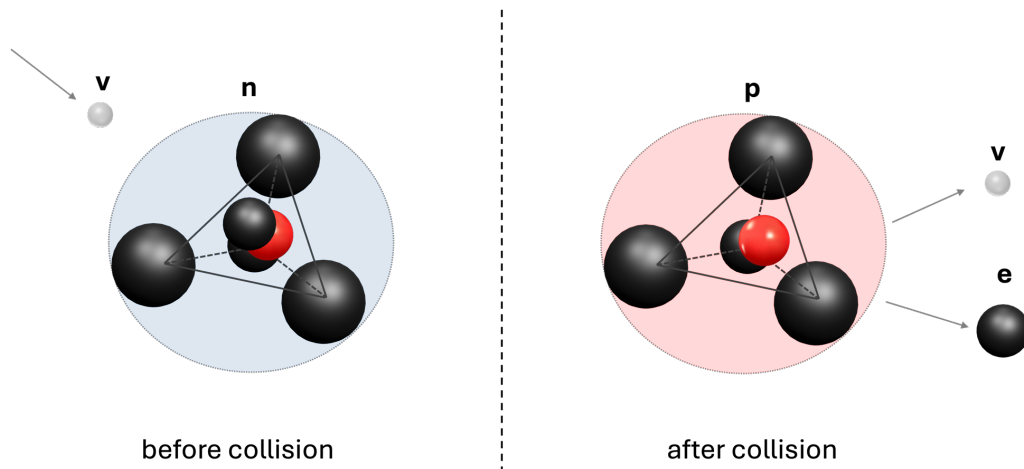


Fig. 5.2 – Neutron Decay using Six Quark to Five Quark Model

6. Atomic Nuclei Decay

Atomic nuclei decay is a fundamental process in nuclear physics, central to our understanding of radioactivity and the stability of atoms. At its core, radioactivity arises from the instability of certain atomic nuclei, which spontaneously undergo decay processes to achieve a more stable configuration. This decay is often accompanied by the emission of

particles and/or electromagnetic radiation. The three most common types of radioactive decay are alpha decay, beta decay, and gamma decay.⁹ Since gamma decay and the release of photons are usually a product of alpha or beta decay occurring first, the latter two types of decays will be discussed in this section.

Beta decay involves the transformation of a neutron into a proton, an electron, and an antineutrino (or neutrino). This is the same process as a free neutron, just now within an atom's nucleus. It typically occurs when the nucleus contains an excess of neutrons. However, unlike the free neutron, the neutron within the nucleus is subject to additional forces and its probability function changes as the requirements to strike the neutron with sufficient force to eject it has now changed. When beta decay occurs, and the neutron decays into a proton, it changes the identity of the nucleus, effectively transmuting one element into another. Beta decay plays a crucial role in the decay chains of many radioactive isotopes, contributing to the natural processes of nuclear transmutation and radioactive dating.

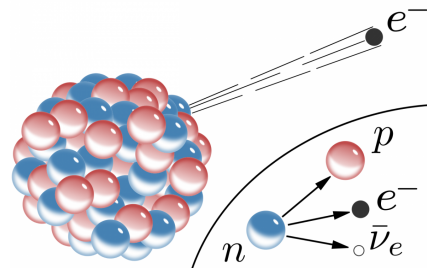


Fig. 6.1 – Beta Decay
Credit: Wikipedia

Alpha decay involves the emission of an alpha particle (helium), which consists of two protons and two neutrons, from the nucleus of an atom. Unlike beta decay, where a neutron converts into a proton, alpha decay involves the ejection of an entire alpha particle from the nucleus. This process is characteristic of heavy and unstable nuclei, where the binding energy per nucleon is not optimal. Alpha decay results in the reduction of the atomic number of the parent nucleus by two and the mass number by four. While alpha decay is different than beta decay, the same cause of beta decay and the decay of a free neutron can be used to explain the emission of an alpha particle. In an atom's nucleus, protons and neutrons are held together by the strong force. A particle, such as the neutrino, randomly striking a proton or neutron may eject an entire alpha particle instead of the quark/electron inside of a neutron.

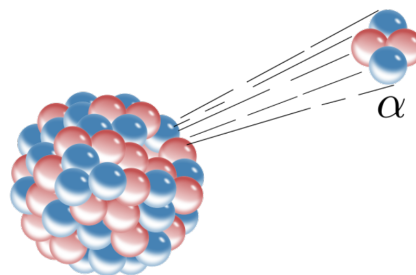


Fig. 6.2 – Alpha Decay
Credit: Wikipedia

While beta and alpha decay processes differ in their mechanisms and products, they share similarities in their underlying probability functions. Just as in free neutron decay, the likelihood of a decay event occurring within a nucleus is influenced by external factors, such as neutrino interactions. Neutrinos can impart sufficient energy to destabilize a nucleus, leading to the ejection of particles through beta or alpha decay mechanisms. Thus, the probability of a neutrino strike affecting the decay process extends beyond free neutrons to include atomic nuclei, underscoring the interconnected nature of nuclear phenomena.

7. Proposed Method to Accelerate Decay

The need to accelerate atomic decay processes arises from various real-world scenarios where the presence of radioactive materials poses risks to human health and the environment. Consider radioactive isotopes with long half-lives, such as uranium-235 (half-life of over 700 million years) or plutonium-239 (half-life of over 24,000 years). These materials are commonly used in nuclear reactors, and their radioactive waste must be stored securely for thousands of years to prevent harm. In the event of a nuclear accident, like the Chernobyl disaster, where radioactive contaminants were released into the environment, affected areas may remain uninhabitable for centuries due to the slow decay of these materials. Accelerating the decay of such radioactive isotopes could significantly reduce the time it takes for contaminated areas to become safe for human habitation.

This paper theorizes that the decay of particles is fundamentally linked to the probability of particles striking the core of an atom's nucleus, with neutrinos being particularly efficient at penetrating the nucleus due to their minuscule size. Therefore, increasing the flux of neutrinos could accelerate decay rates, thereby reducing the half-lives of radioactive isotopes. This raises the intriguing question: could accelerating decay processes be beneficial in rendering radioactive materials stable more rapidly?

Practical applications of accelerated decay processes hold immense promise for humanity. Imagine contaminated sites from nuclear accidents becoming safe for habitation within years rather than centuries. Accelerated decay could revolutionize nuclear waste management, minimizing the need for long-term storage facilities and mitigating environmental risks associated with radioactive materials. Additionally, medical treatments involving radioactive isotopes, such as cancer therapies, could benefit from accelerated decay processes, allowing for shorter-lived isotopes with reduced radiation exposure to patients.

One proposed method to accelerate particle decay involves the manufacturing and targeting of neutrinos towards specific atomic elements that one wishes to expedite the decay of. By generating beams of neutrino particles and directing them towards the desired material, the probability of neutrino strikes increases, thereby accelerating the decay process. A notable example of such a neutrino beam facility is the Neutrinos at the Main Injector (NuMI) at Fermilab.¹⁰ NuMI accelerates protons to high energies, collides them with a target to produce muons and pions, and after further decay eventually produces a beam of neutrinos aimed at detectors for scientific research of the neutrino particle. It is proposed that a similar concept of a neutrino beam could be applied to accelerate decay rates of radioactive isotopes.

To summarize, the proposed method involves the **creation of a neutrino beam that directs neutrinos towards radioactive elements**, effectively speeding up the decay process. By reducing the half-lives of radioactive materials, this approach offers a promising solution to make radioactive waste safer for humans in a fraction of the time required by natural decay processes.

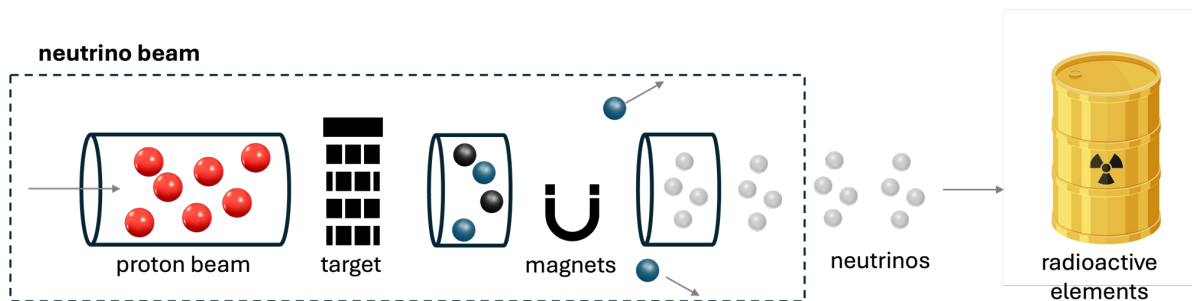


Fig. 7.1 – Neutrino Beam and Radioactive Elements

Conclusion

In conclusion, this paper has explored the relationship between neutron decay and its profound implications for radioactivity. The study of free neutron decay serves as a foundational pillar in comprehending the decay processes of atomic elements, particularly those that exhibit radioactivity. Neutrons play a pivotal role in beta decay, and this paper proposes that the same underlying mechanism responsible for neutron decay may also drive alpha decay processes, thus connecting various facets of nuclear decay phenomena.

The reevaluation of the currently accepted three-quark model of the neutron to a proposed six-quark model, alongside the corresponding adjustment of the proton's model to a five-quark structure, offers a new perspective on the decay mechanisms within atomic nuclei. While this proposed model represents a departure from conventional understanding, it aligns closely with empirical observations of decay products and the mass characteristics of the neutron. Moreover, the proposed probability framework for neutrino interactions with neutrons sheds light on the observed variability in decay rates, particularly concerning the proximity of neutrons to solar events. This probabilistic approach provides a coherent explanation for experimental observations and underscores the intricate interplay between solar phenomena and nuclear decay processes.

While the proposed neutron model and associated decay processes proposed in this paper remain theoretical, they pave the way for experimental validation and potential real-world applications. The proposition of a neutrino beam capable of accelerating the decay of radioactive elements holds immense promise for humanity. By hastening the decay of radioactive isotopes, such a technology could expedite the remediation of contaminated sites, reduce the duration of hazardous radioactive waste storage, and enhance the safety of nuclear materials in medical and industrial applications.

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